

Evidential Mixture-of-Experts for Sentiment Regression

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ABSTRACT

Multimodal sentiment analysis aims to understand human emotions by integrating text, audio, and visual information. Existing methods often struggle with noisy or missing modalities and rarely account for the reliability of each modality. In this paper, we propose E-MoE, an evidence-aware mixture-of-experts framework that models sentiment prediction as an evidential inference problem. Specifically, we first employ modality-specific experts to estimate positive and negative evidences for each modality, capturing both sentiment tendencies and uncertainty. Then, an uncertainty-driven fusion mechanism aggregates these evidences, assigning higher weight to reliable modalities while mitigating the impact of noisy or missing inputs. Finally, the aggregated evidences parameterize a Beta distribution from which the final sentiment prediction is derived. Extensive experiments on CMU-MOSI and CMU-MOSEI demonstrate that E-MoE outperforms state-of-the-art approaches, effectively integrating heterogeneous modalities while providing interpretable and robust sentiment predictions.

CCS CONCEPTS

• Computing methodologies → Computer vision.

KEYWORDS

Sentiment regression, Evidential learning, Beta distribution, Mixture-of-Experts

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1 INTRODUCTION

Recent advances in multimedia technologies and the proliferation of online video content have stimulated growing interest in multimodal video understanding. Videos inherently contain rich

and complementary information across visual, audio, and textual modalities, jointly encoding semantic and event-related cues. This richness has motivated a range of research tasks, including audio-visual understanding [3, 7], dense video captioning [21, 27], audio-visual question answering [1, 8], and multimodal sentiment analysis [6, 10, 20]. In particular, multimodal sentiment analysis seeks to uncover fine-grained emotional states by modeling subtle cues such as facial expressions, vocal patterns, and linguistic semantics, moving beyond early approaches that relied solely on textual signals. Developing robust fusion strategies that effectively integrate complementary modalities while handling noise and uncertainty is therefore essential for accurate and comprehensive emotion understanding in videos.

Multimodal sentiment analysis is commonly formulated as a sentiment score regression task, where the objective is to predict a continuous-valued sentiment intensity for a given video. To this end, existing approaches typically employ deep neural networks to encode different modalities and fuse them using early fusion [16], late fusion [26], or attention-based mechanisms [13]. Recent advances have explored sophisticated cross-modal interaction modeling, including transformer-based architectures [17] and graph-based fusion strategies [11], achieving notable performance improvements. Building on cross-modal attention in transformer-style fusion, alignment-aware variants have been proposed for asynchronous modalities [9]. Distillation with weak affective cues has also been considered [25]. Despite these advances, most existing methods treat sentiment regression as a deterministic prediction problem, focusing primarily on improving point estimation accuracy while largely overlooking the intrinsic uncertainty and ambiguity present in emotional expressions.

A fundamental challenge in sentiment regression lies in the subjective and ambiguous nature of sentiment intensity. Emotional responses are inherently personal, and sentiment annotations often exhibit significant inter-annotator disagreement, particularly for samples with subtle or neutral expressions. However, most existing regression-based methods assume a single ground-truth sentiment value and penalize deviations from this value uniformly, implicitly treating all predictions as equally certain [6, 18, 22]. This modeling assumption is suboptimal, as it fails to account for the fact that some sentiment samples are intrinsically ambiguous and should be associated with higher predictive uncertainty. Motivated by this observation, we argue that sentiment regression should go beyond point estimation and instead explicitly model the strength of evidence supporting different emotional tendencies.

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Another key challenge in multimodal sentiment analysis arises from the inherent inconsistency across modalities. In real-world videos, different modalities may convey conflicting emotional cues. For example, sarcastic expressions may pair positive textual semantics with negative paralinguistic cues, which has motivated sarcasm-aware modeling and shown clear benefits for video sentiment prediction [2]. Meanwhile, many fusion approaches still enforce implicit alignment or consistency across modalities [4, 14, 15], which may lead to overconfident predictions even when modalities disagree. In such cases, it is desirable for the model to express increased uncertainty rather than forcing a deterministic sentiment score. This motivates the need for a modeling framework that can explicitly reason about conflicting multimodal evidence and adaptively weigh modality contributions based on their reliability.

Despite the prevalence of uncertainty induced by subjective annotations and multimodal inconsistency, most existing sentiment regression models do not explicitly estimate predictive uncertainty. Instead, uncertainty is often implicitly absorbed into regression errors, offering limited interpretability and robustness. In this work, we address these limitations by reformulating multimodal sentiment regression from an evidential modeling perspective. We interpret sentiment scores as the expected value of a latent variable representing positive sentiment propensity and model its distribution using a Beta distribution. To this end, we propose an evidential mixture-of-experts framework that learns modality-specific positive and negative sentiment evidences and fuses them to jointly predict sentiment intensity and uncertainty. The main contributions of this work are summarized as follows:

- We propose a novel evidential formulation for multimodal sentiment regression, which models sentiment scores as the expectation of a Beta distribution, enabling unified prediction of sentiment intensity and uncertainty.
- We introduce a mixture-of-experts architecture that explicitly learns positive and negative sentiment evidences from visual, audio, and textual modalities, allowing robust fusion under multimodal inconsistency.
- Extensive experiments on benchmark datasets demonstrate that the proposed method achieves improved regression performance while providing meaningful uncertainty estimates compared to existing deterministic approaches.

2 METHODOLOGY

2.1 Problem Definition

Given a video, three heterogeneous modalities are observed: a visual sequence $V = \{v_t\}_{t=1}^{T_v}$, an audio sequence $A = \{a_t\}_{t=1}^{T_a}$, and a textual sequence $S = \{s_t\}_{t=1}^{T_s}$, with lengths T_v , T_a , and T_s , respectively. Let $m \in \{v, a, s\}$ denote the modality index, and $\mathbf{X}_m$ the corresponding input sequence, i.e., $\mathbf{X}_v = V$, $\mathbf{X}_a = A$, and $\mathbf{X}_s = S$. Each video is associated with a continuous-valued sentiment annotation $y \in [0, 1]$ representing the overall sentiment intensity of the video. The goal of multimodal sentiment analysis is to learn a mapping function:

$$f : (V, A, S) \rightarrow \hat{y}, \quad (1)$$

such that the predicted sentiment score $\hat{y}$ approximates the ground-truth sentiment label y . Existing approaches typically model $f(\cdot)$

as a deterministic regression function and optimize it using a point-wise regression loss (e.g., mean absolute error). In contrast, we argue that sentiment prediction should account for the inherent subjectivity and ambiguity of emotional expressions, which motivates a probabilistic treatment.

Instead of directly regressing a scalar sentiment value, we reformulate sentiment regression as an evidential inference problem. Specifically, we assume that the observed sentiment score y is generated from a latent random variable $p \in [0, 1]$, representing the propensity of the sentiment being positive. We model this latent variable using a Beta distribution:

$$p \sim \text{Beta}(\alpha, \beta), \quad (2)$$

where $\alpha > 0$ and $\beta > 0$ denote the shape parameters. Under this formulation, the predicted sentiment score is given by the expectation of the Beta distribution:

$$\hat{y} = \mathbb{E}[p] = \frac{\alpha}{\alpha + \beta}, \quad (3)$$

while the uncertainty of the prediction is naturally captured by the variance:

$$\text{Var}(p) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}. \quad (4)$$

Following evidential learning and subjective logic principles, we interpret the Beta parameters as accumulated evidences supporting two competing sentiment tendencies:

$$\alpha = e^+ + 1, \quad \beta = e^- + 1, \quad (5)$$

where e^+ and e^- denote the amounts of evidence supporting positive and negative sentiment, respectively. This evidential formulation separates sentiment intensity from predictive confidence: the relative magnitude of e^+ and e^- determines the predicted sentiment score, while their sum controls the model's confidence.

2.2 Modality-Specific Evidential Experts

To extract sentiment-related evidence from heterogeneous modalities, we employ a mixture-of-experts architecture in which each modality $m \in \{v, a, s\}$ is equipped with two experts: one estimating positive evidence e_m^+ and the other estimating negative evidence e_m^- . Given the input sequence of modality m , a modality-specific encoder first maps it into a compact representation:

$$\mathbf{x}_m = \mathcal{E}_m(\mathbf{X}_m), \quad (6)$$

where $\mathcal{E}_m(\cdot)$ denotes the encoder for modality m . The specific architectural instantiations of $\mathcal{E}_m(\cdot)$ are modality-dependent and are detailed in the experimental settings. The modality feature is then mapped to non-negative evidences through lightweight neural networks:

$$e_m^+ = \text{softplus}(f_m^+(\mathbf{x}_m)), \quad e_m^- = \text{softplus}(f_m^-(\mathbf{x}_m)), \quad (7)$$

where the $\text{softplus}(\cdot)$ ensures that the estimated evidences are non-negative. These evidences are further used to parameterize a Beta distribution for each modality:

$$p_m \sim \text{Beta}(\alpha_m, \beta_m), \quad \alpha_m = e_m^+ + 1, \quad \beta_m = e_m^- + 1, \quad (8)$$

from which the expected sentiment score and associated predictive uncertainty are computed as

$$\hat{y}_m = \frac{\alpha_m}{\alpha_m + \beta_m}, \quad u_m = \frac{\alpha_m \beta_m}{(\alpha_m + \beta_m)^2(\alpha_m + \beta_m + 1)}. \quad (9)$$

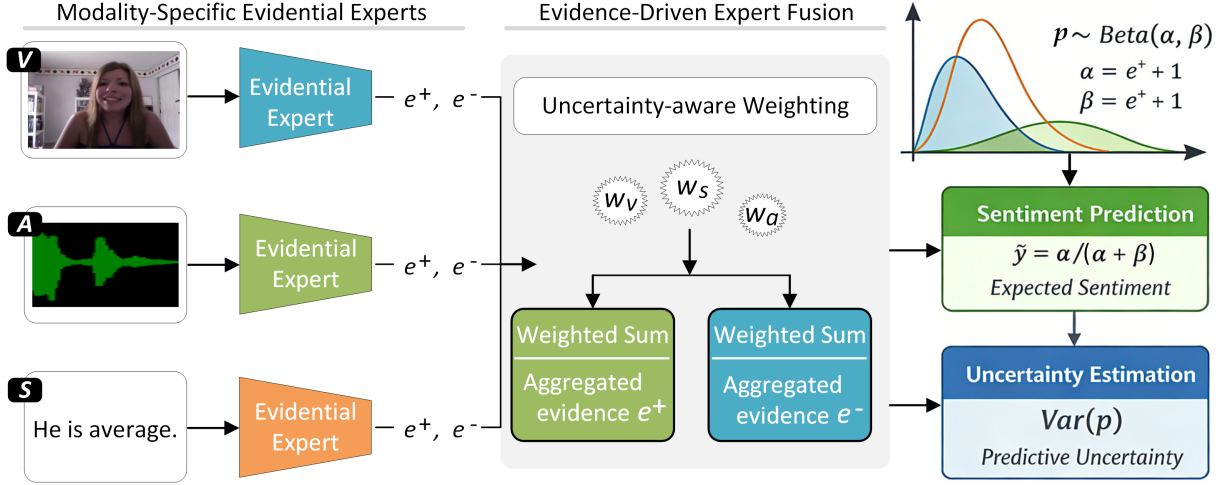


Figure 1: Architecture of the E-MoE. Modality-specific encoders extract features, evidential experts estimate sentiment evidence, and uncertainty-aware fusion produces the final sentiment score.

This design enables each modality to independently quantify the strength of evidence supporting positive or negative sentiment tendencies. Importantly, the experts do not directly predict sentiment scores; instead, they focus on capturing the evidential support derived from modality-specific cues, such as facial expressions in the visual modality, prosodic patterns in the audio modality, and affective semantics in the textual modality.

2.3 Evidence-Driven Expert Fusion

Intuitively, a modality that accumulates strong and consistent evidence should exert greater influence on the final prediction, whereas a modality with high uncertainty should be down-weighted. Based on this observation, we derive modality weights directly from their evidential uncertainty:

$$w_m = \frac{1}{u_m + \epsilon}, \quad \hat{w}_m = \frac{w_m}{\sum_k w_k} \quad (10)$$

where ϵ is a small constant for numerical stability. Unlike conventional mixture-of-experts models that rely on an additional gating network, our weighting mechanism is fully determined by the evidential outputs of the experts themselves, enabling a principled and parameter-free fusion strategy.

The video-level evidences are obtained via weighted aggregation:

$$e^+ = \sum_m \hat{w}_m e_m^+, \quad e^- = \sum_m \hat{w}_m e_m^- \quad (11)$$

These aggregated evidences define the final Beta distribution:

$$p \sim \text{Beta}(e^+ + 1, e^- + 1) \quad (12)$$

from which the predicted sentiment score is computed as:

$$\hat{y} = \mathbb{E}[p] = \frac{e^+ + 1}{e^+ + e^- + 2} \quad (13)$$

This formulation enables unified modeling of sentiment intensity and predictive uncertainty, while allowing each modality expert to influence the final decision in proportion to its evidential reliability.

2.4 Training Objective

The model is trained by maximizing the likelihood of the ground-truth sentiment score under the predicted Beta distribution. Specifically, we minimize the negative log-likelihood (NLL) loss:

$$\mathcal{L}_{NLL} = -\log \text{Beta}(y \mid \alpha, \beta) \quad (14)$$

To discourage the model from generating excessive evidence when predictions are inaccurate, we introduce an evidence regularization term:

$$\mathcal{L}_{reg} = \lambda (y - \hat{y})^2 (e^+ + e^-) \quad (15)$$

where λ is a weighting hyperparameter.

The final training objective is defined as:

$$\mathcal{L} = \mathcal{L}_{NLL} + \mathcal{L}_{reg} \quad (16)$$

This objective encourages the model to accumulate strong evidence only when predictions are reliable, while maintaining high uncertainty for ambiguous or conflicting samples.

3 EXPERIMENTS

3.1 Experimental Setup

Datasets. We evaluate E-MoE on two widely used benchmark datasets, CMU-MOSI [23] and CMU-MOSEI [24]. CMU-MOSI consists of 2,199 annotated utterances extracted from 93 YouTube movie review videos. CMU-MOSEI is a large-scale extension of CMU-MOSI, comprising 23,453 utterances from 3,228 videos, and adopts the same sentiment annotation protocol, providing a substantially larger and more diverse benchmark for evaluation.

Implementation Details. All experiments are conducted on a single RTX 3090 GPU using CUDA 11.6 and PyTorch 1.14. Text, visual, and audio features are extracted using CMU-MultimodalSDK, Facet3, and COVAREP tools, respectively, with input dimensions of (768, 20, 5) for CMU-MOSI and (768, 35, 74) for CMU-MOSEI. For modality-specific feature extraction, a learnable BERT is used for text, while two separate LSTMs are employed for visual and audio features. Consistent with previous studies, standard evaluation

Table 1: Performance comparison on CMU-MOSI and CMU-MOSEI. The left side of “/” is “negative/non-negative” and the right is “negative/positive”. Best results are shown in bold.

Dataset	CMU-MOSI					CMU-MOSEI				
Metric	Acc-2 (↑)	F1 (↑)	Acc-7 (↑)	MAE (↓)	Corr (↑)	Acc-2 (↑)	F1 (↑)	Acc-7 (↑)	MAE (↓)	Corr (↑)
MISA [6]	81.8/83.4	81.7/83.6	42.3	0.783	0.761	83.6/85.5	83.8/85.3	52.2	0.555	0.756
MAG-BERT [17]	82.5/84.3	82.6/84.3	-	0.731	0.789	83.8/85.2	83.7/85.1	-	0.539	0.753
Self-MM [22]	84.0/86.0	84.4/86.0	-	0.713	0.798	83.8/85.2	83.7/85.1	-	0.539	0.753
MMIM [5]	84.1/86.1	84.0/86.0	46.7	0.700	0.800	82.2/86.0	82.7/85.9	54.2	0.526	0.772
GLoMo [28]	84.1/86.7	83.9/86.6	48.3	0.718	0.782	83.7/86.5	84.0/86.4	55.0	0.539	0.771
PFAD [12]	84.6/86.3	84.5/86.2	49.0	0.692	0.799	84.8/86.6	84.6/86.5	55.1	0.525	0.789
DAEV [19]	84.4/86.3	84.5/86.3	46.3	0.730	0.787	83.3/86.1	82.9/86.2	52.3	0.541	0.769
E-MoE	84.8/86.9	84.6/86.9	48.1	0.688	0.804	84.7/86.8	84.8/86.7	55.4	0.525	0.792

metrics are adopted to assess both regression and classification performance, including MAE and Pearson correlation (Corr) for regression, as well as Acc-2, F1 score, and Acc-7 for classification under standard sentiment intervals. The model is optimized using Adam with an initial learning rate of 5×10^{-5} for BERT, 5×10^{-3} for the visual and audio feature extractors, and 1×10^{-3} for the remaining parameters, with the hyperparameter λ set to 0.5. The batch sizes are set to 64 and 32 for CMU-MOSI and CMU-MOSEI, respectively.

Baselines. Comparison is conducted with several baselines: 1) MISA [6], which learns modality-invariant and modality-specific representations. 2) MAG-BERT [17], which incorporates multi-modal inputs into pretrained language models. 3) Self-MM [22], which jointly trains unimodal and multimodal tasks with self-supervision. 4) MMIM [5], which maximizes cross-modal mutual information. 5) GLoMo [28], which exploits global and local representations via mixture-of-experts fusion. 6) PFAD [12], which performs progressive fusion based on audio-visual affective descriptions. 7) DEVA [19], which performs text-guided progressive fusion by converting audio-visual signals into sentiment descriptions.

3.2 Performance Evaluation

Table 1 reports the performance comparison on the CMU-MOSI and CMU-MOSEI datasets. Overall, E-MoE consistently achieves the best or highly competitive results across both datasets and all evaluation metrics. On CMU-MOSI, E-MoE attains the highest Acc-2 and F1 scores while also achieving the lowest MAE and the highest Pearson correlation, indicating more accurate and stable sentiment regression compared with prior methods. Although PFAD achieves the best Acc-7 on CMU-MOSI, E-MoE remains competitive, demonstrating a balanced advantage across both classification and regression metrics. On the larger and more challenging CMU-MOSEI dataset, E-MoE further shows clear superiority, outperforming all baselines in Acc-2, F1, Acc-7, and Corr, and matching the best MAE. These results validate the effectiveness of the proposed E-MoE framework in robustly modeling multimodal sentiment, particularly in leveraging complementary multimodal evidence and reducing prediction uncertainty across diverse evaluation settings.

Table 2: Performance vs. missing rate on CMU-MOSI.

	CMU-MOSI				
	0.1	0.2	0.3	0.4	0.5
Self-MM	81.62	75.47	73.39	70.58	62.47
MMIM	83.04	78.58	76.46	74.53	68.89
E-MoE	84.26	80.72	77.95	75.76	70.83

3.3 Ablative Analysis

Table 2 reports the F1 scores of different methods on the CMU-MOSI dataset under varying data missing rates. As the missing rate increases from 0.1 to 0.5, the performance of all methods degrades, indicating the increased difficulty of sentiment prediction with incomplete multimodal inputs. Nevertheless, E-MoE consistently outperforms Self-MM and MMIM across all missing rates, demonstrating stronger robustness to missing data. Notably, the performance gap becomes more pronounced at higher missing rates (e.g., 0.4 and 0.5), suggesting that E-MoE is more effective at leveraging the remaining reliable modality information and mitigating the impact of missing modalities.

4 CONCLUSION

In this paper, we propose E-MoE, a multimodal sentiment analysis framework that introduces an evidence-aware mixture-of-experts to adaptively capture modality-specific contributions and reduce the impact of missing data. Experiments on CMU-MOSI and CMU-MOSEI demonstrate the effectiveness of the proposed approach, highlighting its ability to robustly integrate heterogeneous modalities and improve sentiment prediction.

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